

Temporary Pedestrian Accessible Routes (TPAR) Manual:

Access for People with Disabilities

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Eugene

SZS ENGINEERING

TEMPORARY PEDESTRIAN ACCESSIBLE ROUTES (TPAR) MANUAL:

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Their combined experience and knowledge helped to refine best practices for temporary pedestrian accessible routes to ensure access through work zones for those in our community who may experience a disability at any point in their lives.

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I. Overview/Background

The Americans with Disabilities Act (ADA) is a federal law that prohibits discrimination against people with disabilities in everyday activities.[1] The ADA contains specific requirements for state and local governments to ensure equal access for people with disabilities. The law requires accommodation for pedestrians with disabilities not only in completed, publicly accessible facilities but also during times of construction or improvement. Another federal law, the Architectural Barriers Act (ABA), requires public agencies to provide access to facilities designed, built, altered, or leased with Federal funds.[2] Sidewalks are considered a facility under federal law.

When temporary traffic control (TTC) zones affect the movement of pedestrians, adequate pedestrian access and walkways shall be provided (Manual on Uniform Traffic Control Devices (MUTCD) (11th Edition, Dec. 2023) Section 6C.02 Pedestrian Considerations, Standard 03). All pedestrians, including the disabled, should be provided with safe, accessible, and reasonably convenient paths that replicate, as nearly as practical, the most desirable characteristics of the existing pedestrian circulation facilities when TTC zones are planned and TPAR is set up (MUTCD Section 6C.02 Pedestrian Considerations, Guidance 07).

This document, **Temporary Pedestrian Accessible Routes (TPAR) Manual: Access for People with Disabilities** is designed for project managers, designers, contractors, utility companies, tree trimming or painting companies, field inspectors, and anyone else involved with the design or implementation of TPAR. The intent of this manual is to describe TPAR requirements, methods, and best practices to ensure accessibility for people with disabilities in work zones, including those who are blind or have low vision.

For purposes here, the definition of work in public rights-of-way refers to alterations of existing elements, new construction, maintenance, repairs, or similar that cause interruptions in the continuous unobstructed path connecting accessible elements and spaces. Work zones are areas of a roadway or travel way where construction, maintenance, or utility work activities take place. A work zone extends between the first warning sign to the last TTC device.

The needs of pedestrians with disabilities must be addressed in temporary traffic control plans (TCPs) in accordance with the ADA. Designers, planners, and contractors creating and implementing TCPs have two primary considerations: design for the built environment and communications.

1. **Design for the built environment:** physical devices, measurable components such as distances and slopes and their established design parameters, and maintaining access (i.e. TPAR connectivity) to businesses, transit, and other nearby destinations. The TPAR must provide a safe detour for all users around the work zone. Worksite personnel must ensure that the TPAR is adequately maintained and pedestrian walkways are kept clear throughout the duration of a project.

2. **Communications:** Rules issued by the US Department of Justice under the ADA require local governments to ensure that communications with individuals with disabilities are as effective as communications with others.[3] Examples of effective communication include, advising pedestrians of potentially hazardous conditions in sidewalks and paths of travel associated with work zones before

[1] ADA, 42 U.S.C. § 12101 et seq.

[2] ABA, 42 U.S.C. § 4151 et seq.

[3] 28 C.F.R. § 35.160 - General

they encounter them, and the use of appropriate auxiliary aids onsite for people with disabilities. TPAR information for capital improvement projects provided on the City's website shall be provided in a format following Web Content Accessibility Guidelines (WCAG) 2.1 Level A and Level AA Success Criteria.[4]

TPAR Considerations for People with Disabilities

Engineers, architects, contractors, and building officials must keep in mind pedestrians with disabilities, including those with vision and mobility disabilities. According to 2024 data from the Centers for Disease Control and Prevention (CDC), more than 1 in 4 adults in the US (28.7%) have some type of disability.[5] Around 30% of the US population does not drive.[6] People with disabilities live, work, and navigate throughout our communities every day. Many disabilities can impact how a pedestrian may be able to navigate a work zone safely.

Some people may have more than one disability, which may or may not be apparent to others. For example, a person who is blind or has low vision may also have a hearing impairment, mobility impairment, or cognitive impairment. Since people with different disabilities have different needs, circumstances on a work site often must rely on judgment.

Consideration for People with Vision Disabilities

According to CDC data from 2024, 5.5% of adults in the US have a vision disability, described by the CDC as blindness or serious difficulty seeing.[7] People who are blind or have low vision are not a homogeneous group. There is significant variation across individuals regarding their degree and type of vision loss.

Safety Concerns for People with Vision Disabilities

A small percentage of people with vision disabilities have guide dogs that lead them away from or around hazards.[8] Those without guide dogs must find their own way, whether or not there is a safe route for them to use. When a person with a vision disability finds a safe path, they may use it exclusively to remain safe while traveling, which makes changes caused by work zones a safety risk. When work zones do not provide TPAR, pedestrians may travel into the roadway which can be dangerous for any pedestrian, especially those with vision disabilities.

- Common TPAR wayfinding challenges and safety issues for pedestrians with vision disabilities are:
- Difficulty recognizing that they have come upon a work zone.
- Not knowing if there is a safe alternative way to proceed when a sidewalk is closed.

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[4] 28 C.F.R. § 35.200 - Requirements for web and mobile accessibility

[5] CDC, Disability Impacts All of Us Infographic (July 15, 2024)

[6] Federal Highway Administration (FHWA), Licensed Drivers, Vehicle Registrations, and Resident Population (In Millions), Highway Statistics 2023 (March 11, 2025)

[7] CDC, Disability Impacts All of Us Infographic (July 15, 2024)

[8] Neoma Stark, How many blind people use dogs? A Complete Guide, KeepingDog (February 1, 2023)

- Difficulty determining if they should attempt to cross the street and continue by walking on the other side or if they should retrace their steps and use another route.
- Challenges accessing businesses within the sidewalk closure area.
- Trouble finding a bus stop if it has been relocated.
- Not knowing if detectable edging might have sharp or abrasive edges that could injure someone feeling their way along by touch.
- Not knowing if a white cane can help them find a safe way to continue through the detour.
- Difficulty discerning what type of traffic control is at an unfamiliar intersection.
- Challenges getting back on their familiar route.
- Not knowing if there might be someone nearby who could offer them assistance.

Consider this scenario: A person who is blind is on their way to a job interview. They encounter caution tape and cones within their normal path of travel. They ponder where to go to get to their transit stop. Unsure where to go and finding no one to ask, they return home and use rideshare to assure they do not miss their interview. They arrive only a few minutes late for their interview but are very stressed.

Consider the same scenario but with a different experience: Instead of floundering on caution tape and cones, the TPAR is communicated via audible messages. The TPAR also uses a channelizing barrier that a white cane can detect to move them into and out of the street, with compliant temporary ramps and detectable warning surfaces. The person can continue to their transit stop in plenty of time to catch the bus, arriving at their interview relaxed and a few minutes early, as planned.

In both cases the individual did make their interview, but the first scenario does not communicate that a work zone is ahead nor provide a channelized pedestrian access route to allow them to move around the hazardous work zone. Without the ability to see a detour at a work zone, pedestrians with vision disabilities have significant disadvantages in comparison to the general public.

Design Considerations for People with Vision Disabilities

In designing accessible routes, it is important to consider the needs of pedestrians with vision disabilities to ensure everyone is kept as safe as possible. People with vision disabilities need a physical environment that is free of sharp edges, uneven grading, obstructions that can cause injury, and vertical or horizontal changes in the walking surface that can cause tripping or falling hazards.

Those with vision disabilities may not be able to recognize where the street begins when curbs are flush with the sidewalks. Detectable warning surfaces are tactile material installed before the flush curb in a blended transition or curb ramp. Detectable warning surfaces consist of truncated domes in a square or radial grid pattern (Public Right-of-Way Accessibility Guidelines (PROWAG) R305.1 General). At the boundary between the sidewalk and vehicle travel lane, detectable warning surfaces can be felt by a person navigating with a white cane or underfoot by a person with a dog guide.

Some people with low vision may depend on colors as a warning. Detectable warning surfaces shall visually contrast with the walking surface (light-on-dark or dark-on-light) (PROWAG R305.1.3 Contrast) so that people with low vision can recognize the detectable warning before they enter the street.

Communicating a TPAR for People with Vision Disabilities

A city can ensure accessibility by improving communication at the site of the work zone. Visual signs are a typical part of TPAR but are not usable by pedestrians with vision disabilities. Blocked routes, alternate crossings, and sign and signal information should be communicated to pedestrians with vision disabilities by providing audible information, tactile and/or vibrating surface devices, accessible pedestrian signal walk indications, and barriers and channelizing devices with continuous detectable edging (PROWAG R303.2 Signs, R307 Pedestrian Push Buttons and Passive Pedestrian Detection, R308 Accessible Pedestrian Signal Walk Indications, R303.6 Detectable Edging of Channelizing Devices, and MUTCD Section 6C.03 Accessibility Considerations, Guidance 06).

Audible information devices (AIDs) described in the PROWAG indicates that signage that directs pedestrians to and from detours shall be provided via proximity-actuated (vs. button push) audible signs or other non-visual format so that it is accessible to people with vision disabilities (PROWAG R303.2 Signs). Commercially available AIDs can provide the required audible signs using infrared or other motion sensors. Since AIDs are known to be susceptible to theft, the devices should be secured to the traffic control device (barricade, channelization, fence, etc.).

The audible message can be customized so the devices can be re-used for multiple construction projects over time. The audible message should be brief, to the point, spoken clearly, and recorded in a quiet area. Incorporate directions that do not require visual recognition. Message example: “Attention eastbound Washington Avenue pedestrians. You are at the southeast corner of Washington and Church. Road construction on Washington from Church St to Huron for 7 blocks. Protected pedestrian path open on this side. Work will continue until August 20.”[9]

AIDs may not be needed if detectable channelizing devices make an alternate route of travel evident to pedestrians with vision disabilities (MUTCD Section 6C.03 Accessibility Considerations, Standard 04, Support 05).

Where field personnel are available to guide pedestrians through work zones (see MUTCD Section 6C.02 Pedestrian Considerations, Option 04-05), best practices for assisting a person with a vision disability are as follows:

- Ask if they want to follow you or take your arm (sighted guide).
- To guide, if they do not take hold of your elbow, with the back of your hand touch their lower arm and say, “Do you want my arm?”
- If they have a dog, guide from the opposite side from the dog.
- Walk at their pace.

[9] Minnesota Department of Transportation, Temporary Pedestrian Access Routes (TPAR) Audible Message Content (January 2018)

- If they are following you, keep up the conversation so they know where you are.

Consideration for People with Mobility Disabilities

According to CDC data from 2024, 12.2% of adults in the US have a mobility disability, described by the CDC as serious difficulty walking or climbing stairs.[10] People with mobility disabilities may use canes, crutches, walkers, wheelchairs, scooters, or no assistive devices at all. Respiratory, circulatory, and neurological medical conditions can cause mobility disabilities that most people will not notice. Field personnel in work zones should not rely on visual cues that someone might have a disability, and should expect to always maintain TPAR, as legally required.

Safety Concerns for People with Mobility Disabilities

Common safety issues for pedestrians with mobility disabilities are:

- Temporary ramps that are too steep, wobble, or do not have an adequate side guard rail to grasp.
- Pathways and turns or turning spaces that are too narrow for wheelchairs, motorized scooters, or those who travel with service animals.
- Ramps or pathways that lack adequate edge protection along the outer edges, which during wet weather can allow mud or debris to flow over the surface of the ramp or path, creating slippery conditions that can be very dangerous to those who use crutches, canes, or walkers.

Design Considerations for People with Mobility Disabilities

To ensure accessibility at work zones, it is important to consider the needs of pedestrians with mobility disabilities. A person with a mobility disability may have difficulty using steps or vertical curbs, or navigating narrow pathways, turnarounds, severe slopes, or changes in elevation. These components of the work zone can prevent travel, create tripping hazards, or cause individuals to fall out of their mobility assistance devices. Even small elevation changes are sometimes difficult for people with mobility disabilities to traverse, which is one reason that installing ramps is essential to maintaining access.

People using wheelchairs or other mobility devices rely on flush curbs or transitions to safely continue travel. For a pedestrian with a mobility disability, a vertical barrier acts like a wheel stop does for a vehicle. Detectable warning surfaces that consist of truncated domes in a square or radial grid pattern allow wheeled devices to travel in the flat troughs between them. For accessible TPAR design, the needs of pedestrians with mobility disabilities must be considered.

II. Sources

Laws, regulations, and understanding of how people with disabilities access the built environment have changed over the last few decades. Guidance that applies to TPAR has also changed. This manual relies primarily on the most recent statutes, regulations, and research on usability for pedestrians with disabilities. (See cover for date of publication.) This manual also uses guidance from state and federal agencies, as well as organizations that perform research on pedestrian use of public rights-of-way for people with disabilities. This manual is meant to supplement the document Pedestrian Accommodation in Work Zones: A Field Guide

[10] CDC, Disability Impacts All of Us Infographic (July 15, 2024)

developed by the American Road & Transportation Builders Association Work Zone Safety Consortium. (Terms such as “guidelines,” “practices,” “requirements,” and “standards” are used throughout this manual. The use of these or similar terms is not intended to create a cause of action.)

Please Note: Many of the documents referencing TPAR available online reflect requirements from outdated versions of federal sources such as older versions of the Manual on Uniform Traffic Control Devices (MUTCD), the 1994 Americans with Disabilities Act Accessibility Guidelines (ADAAG), and the 2005 Public Right-of-Way Accessibility Guidelines (PROWAG) from the US Access Board. Do not rely on outdated sources that have changed substantially since they were originally developed.

The Settlement Agreement resulting from Moore et al. v. City of Eugene et al.[11] has established applicable sources to govern this manual:

- Manual on Uniform Traffic Control Devices (MUTCD): The currently adopted federal document setting minimum standards, supplemented with guidance, options, and support for traffic control across the US. The Federal Highway Administration has required all state highways and public roadways under the jurisdiction of cities and counties to conform to the MUTCD.[12] Oregon adopted the MUTCD and an Oregon Supplement.[13] Pursuant to Eugene Code 7.007, the City Engineer adopted the use of the MUTCD in both the City of Eugene Public Improvements Design Standards (PIDS) Manual and the City of Eugene Standard Specifications for Construction (COE SSC). MUTCD Section 6 is the chapter most applicable to TPAR. The 11th Edition of the MUTCD (December 2023) is the most current version published by the Federal Highway Administration and is the version cited throughout this manual.

- Public Right-of-Way Accessibility Guidelines (PROWAG): Federal accessibility guidelines that provide scoping and technical standards for ensuring that sidewalks, crosswalks, shared-use paths, pedestrian signals, on-street parking, and other pedestrian facilities located in the public right-of-way (space that forms the boundary of a site or that lies within a site bounded by a property line), are readily accessible to and usable by pedestrians with disabilities (see PROWAG R101.1 Purpose). The PROWAG was issued by the US Access Board on August 8, 2023.[14] These guidelines have been adopted by the Department of Transportation and will be mandatory after they are adopted by the Department of Justice under Title II of the ADA. In lieu of an adopted enforceable federal standard, the City of Eugene has adopted the guidelines administratively in PIDS Manual and the Standard Specifications for Construction. Note: where the PROWAG guidelines are more stringent than the MUTCD Standards, the PROWAG shall control. (See the PIDS Sections 3.05 Temporary Traffic Control Plan and 4.13 Traffic Control Plan.)

- Oregon Department of Transportation (ODOT) Highway Design Manual: Identifies standards for a variety of roadway design features on State Highways.

[11] No. 6:21-cv-01293-MTK, (D. Or. filed Aug. 31, 2021)

[12] 23 C.F.R. § 655.603(a) – National MUTCD

[13] O.R.S. § 810.200 - Uniform standards for traffic control devices; O.A.R. 734-020-0005 - Traffic Control Devices

[14] 36 C.F.R. Part 1190

- Oregon Standard Drawings: Adopted as the standards by ODOT for the State Highway System. The City of Eugene has administratively adopted an amended version of most of the State standard drawings (City of Eugene Standard Drawings (COE SD) TM-Series). Standard Drawing TM844 addresses TPAR.

- City of Eugene Public Improvements Design Standards (PIDS) Manual: Provides design standards for projects within City limits, specifically:

- 3.05 Temporary Traffic Control

- 4.13 Traffic Control

- City of Eugene City Code: Spells out how public right-of-way projects and improvements should be carried out. It includes references to the City's adopted PIDS. Chapter 7 of City Code addresses public improvements, specifically:

- 7.085 Standard Specifications, Drawings and Design Standards

- 7.090 Temporary Work

- 7.307 Public Way Construction and Use - Repair

III. Defined Terms

The terms used in this document have been drawn from the applicable sources. Where necessary, additional commentary has been added for improved clarity.

Accessible: A pedestrian facility or element in the public right-of-way that complies with the PROWAG (PROWAG R104.3 Defined Terms: A). Commentary: The combination of various elements in a facility, site, or area, or portion thereof which allows access, circulation, and the full use of the facilities by people with disabilities.

Audible Information Device (AID, Audible Signage): An audible information device (AID) is a device designed to provide information to pedestrians with vision disabilities. It delivers audible messages equivalent to visual signage for tasks such as notifying pedestrians about sidewalk closures and detours. Commentary: the PROWAG refers to AIDs as proximity actuated audible signs (PROWAG R303.2 Signs).

Blended Transition: Blended transitions are raised pedestrian street crossings, depressed corners, or similar connections between pedestrian access routes at the level of the sidewalk and the level of the pedestrian street crossing that have a grade of 5% or less. Blended transitions are suitable for a range of sidewalk conditions.

Cross Slope: The slope that is perpendicular to the direction of pedestrian travel (PROWAG R104.3 Defined Terms: C). Commentary: The slope of a walking surface is regulated in the two directions of travel (see running slope). Cross slope makes forward travel for people using wheelchairs increasingly difficult. As the cross slope increases, travel becomes increasingly difficult as a wheelchair or walker is pulled in the direction of the slope, risking tipping over and potential injury to themselves or others in their path. Cross slope impedes travel on uphill and downhill slopes by pedestrians who use wheeled mobility devices, braces, lower limb prostheses, crutches, or walkers, as well as pedestrians who have gait, balance, or stamina impairments.

Curb Line: A line at the face of the curb that marks the transition between the curb and the gutter or street (PROWAG R104.3 Defined Terms: C).

Curb Ramp: A sloped connection that is cut through or built up to a curb. Curb ramps may be perpendicular or parallel to the curb or to the street they serve or be a combination thereof (PROWAG R104.3 Defined Terms: C). Perpendicular curb ramps have a running slope that is perpendicular to the curb or the street it serves, and parallel curb ramps have a running slope parallel to the curb or street it serves. (See PROWAG R304 Curb Ramps and Blended Transitions for additional requirements.)

Detectable: Having a continuous edge within 6 inches of the surface so that pedestrians with vision disabilities can sense its presence and receive usable guidance information (MUTCD Section 1C.02 Definitions of Words and Phrases Used in this Manual, No. 58).

Detectable Warning Surface: A standardized surface feature built in or applied to pedestrian circulation paths and other pedestrian facilities to warn of hazards (PROWAG R104.3 Defined Terms: D). Detectable warning surfaces mark boundaries between pedestrian and vehicular ways where there is no raised curb. Detectable warning surfaces are installed where curb ramps are constructed at the junction of pedestrian access routes and the roadway or shoulder, for marked and unmarked crosswalks. Detectable warning surfaces contrast visually with adjacent walking surfaces, either light-on-dark or dark-on-light (MUTCD Section 3C.01 General Crosswalk Markings, Support 04). Detectable warning surfaces shall consist of truncated domes in a square or radial grid pattern (PROWAG R305.1 General). Commentary: Truncated domes mark the junction where pedestrians travel into and out of the roadway.

Disability: With respect to an individual—a physical or mental impairment that substantially limits one or more major life activities of such individual; a record of such an impairment; or being regarded as having such an impairment.[15] Commentary: In the context of the ADA, disability is a legal definition and may vary from other medical or social justice-related definitions.

Diversion or Detour: Alternate pedestrian path of travel around a work zone. (See Temporary Pedestrian Accessible Route.)

Guide Dog (Dog Guide): A guide dog responds to the handler's verbal and/or hand commands (e.g., right, left, forward, hop up, etc.). Guide dogs do not make decisions about the route or decide when to cross a street—these decisions are made by the person they are serving.

Mobility Disability (Physical Disability): Mobility impairment refers to the inability of a person to use one or more of their extremities, or a lack of strength to walk, grasp, or lift objects. A wheelchair, crutches, or a walker may be utilized to aid in mobility. Mobility impairment may be caused by a number of factors, such as disease, an accident, or a congenital disorder and may be the result from neuro-muscular and orthopedic impairments.[16]

ODOT TTCH: Oregon Department of Transportation Temporary Traffic Control Handbook for Operations of Three Days or Less (2026 Edition)

Orientation and Mobility Specialist (COMS, Certified Orientation and Mobility Specialist): A Certified Orientation and Mobility Specialists (COMS) is a trained expert who works with individuals who are blind, have

[15] ADA, 42 U.S.C. § 12102 – Definition of disability

[16] Mobility and Physical Disabilities, Accessibility.com (quoting The ACCESS Project)

low vision, or who have visual limitations. A COMS might evaluate and teach adaptive techniques such as the use of a white cane, guide dog, and/or smartphone apps that help with wayfinding. Through mobility training, people who are blind or have low vision develop skills such as route planning, landmark identification, street crossing, and navigating public transit stops and stations.

Pedestrian: A person on foot, travelling by wheelchair or other mobility device, on skates, or on a skateboard. (PROWAG R104.3 Defined Terms: P). Commentary: An individual who moves in walking areas with or without the use of walking assistive devices such as crutches, leg braces, wheelchairs, white cane, service animal, etc.

Pedestrian Access Route (PAR): An accessible, continuous, and unobstructed path of travel for use by pedestrians with disabilities within a pedestrian circulation path (PROWAG R104.3 Defined Terms: P). Commentary: The PAR is an accessible route in public rights-of-way, or a continuous unobstructed accessible route connecting accessible elements and spaces that can be negotiated by people with a disability using a wheelchair, and that are also safe for and usable by people with other disabilities such as blindness or low vision. PARs may include parking access aisles, curb ramps, crosswalks at vehicular ways, walks, ramps, and lifts with the intent to remain usable for people with disabilities during construction, repair, or maintenance work.

Pedestrian Channelizing Device (PCD): Pedestrian channelizing devices may be used to close a PAR or indicate a suitable path for pedestrian travel around or through the work zone. Devices used to channelize pedestrians shall be detectable to users of white canes and visible to pedestrians with vision disabilities. When used as a sidewalk closure, the device shall cover the entire width of the sidewalk (MUTCD Section 6K.02 Pedestrian Channelizing Devices, Standard 06-07). The MUTCD allows several other devices such as longitudinal channelizing devices and temporary traffic barriers to be used as PCDs under certain conditions. Additionally, the PROWAG and MUTCD provide information on the requirements related to detectable edging of channelizing devices.

Pedestrian Circulation Path: A prepared exterior or interior surface provided for pedestrian use in the public right-of-way (PROWAG R104.3 Defined Terms: P). Commentary: A PAR is a subset of the pedestrian circulation route. In an existing facility constructed prior to the effective date of the ADA, the circulation route may not be accessible in its entirety, but the PAR must be accessible.

Pedestrian Facility: A structure, route, or space for pedestrian circulation or use located in the public right-of-way (PROWAG R104.3 Defined Terms: P).

Pedestrian Hybrid Beacon (PHB): A special type of hybrid beacon used to warn and control traffic at an unsignalized location to assist pedestrians in crossing a street at a marked crosswalk (PROWAG R104.3 Defined Terms: P). PHBs are intentionally placed in a dark mode (no indications displayed) between periods of operation and, when operated, the device displays both steady and flashing traffic control signal indications (MUTCD Section 1C.02 Definitions of Words and Phrases Used in this Manual, No. 105). Commentary: A traffic control device can be used to increase awareness for motorists of pedestrian crossings at uncontrolled marked crosswalk locations. A PHB is only activated by pedestrians when needed.

Positive Protection: Measures taken between work zone field personnel and motorized traffic which contain and/or redirect vehicles and meet applicable crashworthiness criteria.[17] Positive Protection may include a highly mobile barrier, movable and temporary steel barrier, movable concrete barrier, traditional concrete

[17] See 23 C.F.R. § 630.1104

barrier, associated cushions, and other strategies to avoid traffic crashes in work zones, including full road closure. Mobile and other movable barriers enable field crews to quickly create work zones that are physically separated from moving traffic and quickly removed from the roadway once the work is completed.

Rectangular Rapid Flashing Beacon (RRFB): RRFBs consist of two, rectangular-shaped yellow indications, each with a light-emitting diode (LED)-array-based light source (MUTCD Section 4L.02 Design of Rectangular Rapid Flashing Beacons, Standard 01). RRFBs flash with an alternating high frequency when activated to enhance conspicuity of pedestrians at the crossing to drivers.

Running Slope: The slope that is parallel to the direction of pedestrian travel (PROWAG R104.3 Defined Terms: R). Running slope is referred to as longitudinal or running slope. The slope percentage determines whether a walking surface is a sloped walkway or a ramp regulated under the ADA and the PROWAG. Running slope in a pedestrian facility is further defined by location in reference to the roadway.

Temporary: Structures or elements intended for use for a limited amount of time during work. State and federal accessibility standards apply to temporary and permanent structures (see 2010 ADA Standards for Accessible Design (ADAS) 201.3 Temporary and Permanent Structures).

Temporary Curb Ramp (Temporary Ramp): See requirements for Curb Ramps.

Temporary Pedestrian Accessible Route (TPAR): (See PAR for definition of pedestrian access route.) A TPAR is path of travel, marked by signing, delineation, and traffic control devices, used by pedestrians to navigate through or around work zones. The TPAR is included as part of the Traffic Control Plan. A TPAR is required when a PAR is closed due to construction, maintenance operations, or other similar conditions.

Commentary: The PROWAG and MUTCD refer to TPAR as alternate pedestrian access routes. Both the PROWAG and MUTCD require TPAR to be as accessible as the existing PAR. However, the PROWAG permits an alternate means of providing access for pedestrians with disabilities, if establishing or maintaining a TPAR is technically infeasible (PROWAG R204.1 Alternate Pedestrian Access Route). The MUTCD provides options such as construction staff assisting pedestrians through a work zone or providing shuttle service around a work zone (MUTCD Section 6C.02 Pedestrian Considerations, Option 04-05). The PROWAG includes more specific requirements for alternate pedestrian access routes, describing the required level of detail on sidewalk closures and rerouting. In many cases, compliance with the requirements for TPAR will not entail any change from baseline conditions because they closely align with existing MUTCD provisions on temporary traffic control. Additional PROWAG and MUTCD guidance includes the following elements:

- Signs to warn pedestrians of an alternate route in advance to indicate location of detour (MUTCD Section 6I.02 Detour Signs and Plaques and PROWAG 303.2 Signs).

- A smooth, continuous alternate pedestrian route, including curb ramps (or an absence of curbs) (MUTCD Section 6C.02 Pedestrian Considerations, Guidance 11(C)) and PROWAG R303.5 Curb Ramp or Blended Transition).

- Accessible pedestrian signals, AIDs, or detectable barriers and channelizing devices to communicate signal information to travelers with low vision (see MUTCD Chapter 4K Accessible Pedestrian Signals and Detectors, PROWAG R303.7 Pedestrian Signal Heads, R303.2 Signs, and R303.6 Detectable Edging of Channelizing Devices).

- Continuous detectable edging with a bottom surface no higher than 2 inches above the ground and a top surface no lower than 32 inches above the ground (MUTCD Section 6K.02 Pedestrian Channelizing Devices, Standard 08 and PROWAG R303.6 Detectable Edging of Channelizing Devices).

Temporary Pedestrian Accessible Route Plan (TPAR Plan): A written and drawn plan within the TCP that identifies requirements for providing safe, effective, and accessible routes for pedestrians through or around the work zone including TPAR details, advance public notification, and construction and maintenance responsibilities (COE SSC, Section 00110.20).

Temporary Traffic Control Plan (TCP): A TCP describes measures to be used for facilitating road users through a work zone or an incident area. TCPs play a vital role in facilitating road user flow when a work zone, incident, or other event temporarily disrupts normal road user flow. Important auxiliary provisions that cannot conveniently be specified on project plans can easily be incorporated into Special Provisions within the TCP. Provisions for effective continuity of accessible circulation paths for pedestrians through the work zone should be incorporated into the written TCP (MUTCD Section 6B.01, Guidance 11). Note: the MUTCD refers to temporary traffic control plans as TTC plans.

Traffic Control Devices: Signs, signals, markings, and other devices placed on, over, or adjacent to a roadway used to regulate, warn, or guide public traffic by authority of a public body or official having jurisdiction. Commentary: Both temporary and permanent structures are required to comply with federal accessibility requirements (ADAS 201.3 Temporary and Permanent Structures). Many options exist on the market to provide temporary walkways, ramps, and barricades separating pedestrians from traffic.

Vision Disability: Blindness is the inability to see or a lack of vision that cannot be corrected with eyeglasses, contact lenses, eye drops, other medical therapy, or surgery. According to the Social Security Administration, a person who is legally blind has a central visual acuity of 20/200 or less in the better eye with the best correction, as measured on a Snellen vision chart, or a visual field of 20 degrees or less.[18] Approximately 85% of people with visual impairments have some usable vision and may consider themselves low vision or partially sighted rather than blind.[19] Common types of visual impairments include reduced acuity (inability to see details), visual field deficit (reduced total area of what a person can see), contrast sensitivity (ability to detect subtle gradations between the object and its background), night blindness (reduced vision in dim illumination), and glare sensitivity (reduction of visual acuity in bright lighting or other adverse consequences of environmental illumination).

[18] 20 C.F.R. § 404.1581 Meaning of blindness as defined in the law

[19] Cleveland Clinic, Legally Blind (Sept. 2, 2024)

White Cane (Long Cane): A white cane is the most prevalent mobility device used by people who are blind or have low vision. A white cane provides protection and information and detects obstacles or drop-offs in the person's path at ground level. A white cane does not detect obstacles overhanging the path such as branches or signs that protrude horizontally into the pedestrian path. A white cane also identifies the person as an individual who is blind or has low vision to others.

Work Zone: Areas of a roadway or travel way where construction, maintenance, or utility work activities take place. A work zone extends between the first warning sign to the last TTC device.



Source: lhblind.org

IV. TPAR Design Elements & Considerations

Devices and Features

TPAR devices and features that are required to provide an accessible alternate pedestrian pathway are similar to those required to guide traffic through a work zone. As with drivers, pedestrians must be warned in advance of changing conditions and advised of their options for alternate routes with information that they can access. When they approach the work area, pedestrians with disabilities need guidance on any changes from the normal route, such as shifts in the walkway alignment, in the same way that drivers must be warned about lane changes. As with vehicles, when pedestrians traverse the work zone, the route must be clearly marked, have a suitable surface, and have protection from other traffic and work operations, including drop-offs. When pedestrians exit the work zone, they need to be guided back to their original route.

Clear Width

Pedestrians with wheelchairs and other mobility assistance devices need TPAR to be at least 60 inches wide so that two people can pass by each other if they meet head on. If a 60-inch continuous path cannot be created over the entire length, a minimum 48-inch walkway with a 60-by-60-inch passing space every 200 feet maximum shall be provided (see MUTCD Section 6C.02 Pedestrian Considerations, Guidance 11(D)).

Signs and Surface Delineators

Printed signs and vertical surface delineators (cones or tubular markers) are not usable by pedestrians with vision disabilities. Therefore, blocked routes, alternate crossings, and sign and signal information should be communicated to pedestrians with vision disabilities by providing audible information devices, tactile and/or vibrating surface devices, and barriers and channelizing devices that are detectable to pedestrians traveling with the aid of a white cane or who have vision disabilities (MUTCD Section 6C.03 Accessibility Considerations, Guidance 06). Technology today allows the provision of information to pedestrians with vision disabilities that is equivalent to visual signing for notification of sidewalk closures in a speech message provided by an AID. Devices that provide speech messages in response to passive pedestrian actuation are the most desirable, as they do not require actuation.

Walking Surfaces

Surface conditions are required to be firm, stable, and slip-resistant (ADAS 302.1 General and PROWAG R302.6 Surfaces). In practical terms, paths of travel should be clear of debris and adequately drained to prevent the accumulation of trash or mud that can cause a tripping or slipping hazard. The TPAR should be able to support a wheelchair under all conditions (dry, rain, snow, etc.). The TPAR shall be checked daily for any standing water or mud to ensure debris such as sand, dirt, or leaves are not creating a slippery walking surface.

The surface of the pedestrian path should be as clear and as flat as possible. Many pedestrians with mobility disabilities cannot navigate steps or uneven surfaces without becoming disoriented or losing their balance.

The 2010 ADAS require horizontal joints or other openings to be no more than ½ inch wide or flush to prevent tripping and to reduce the possibility of canes or small wheels getting trapped in gaps or open spaces. If drainage grate openings are located within the pedestrian route, openings must be perpendicular to the pathway and must be narrow enough that a sphere greater than ½ inch in diameter will not pass through (see ADAS 302.3 Openings and 403.2 Floor or Ground Surface). Sidewalk ramp surfaces shall have a running slope not to exceed 8.3% and cross slope not to exceed 2.1% (PROWAG R304.2.1 Running Slope and R304.2.2 Cross Slope). Gutter surfaces in the roadway can be problematic when the alternate route is required to be at least as accessible as the existing path that was previously available to pedestrians. Engineering judgment must be used.

Requirements for detectable edging are dependent on the location. Where a channelizing device is used, detectable edging shall be located along the walking surface with openings no higher than 2 inches from the ground with a top edge at least 32 inches high and free of sharp edges (see PROWAG R303.6.1 Top and R303.6.2 Bottom). Detectable pedestrian edging should be orange, white, or yellow and should match the color of the adjacent channelizing or traffic control devices, when present.

Temporary Curb Ramps

When pedestrians are diverted to a TPAR in the roadway and a vertical curb separates the roadway from the sidewalk, temporary curb ramps must be provided to enable pedestrians to safely travel around the work zone. Temporary curb ramps can be constructed out of a variety of materials including asphalt, wood, and synthetic pre-made systems. Each ramp must be regularly maintained to ensure usability for the life of the ramp's use.

Temporary curb ramps shall be the full width of the temporary route, with a recommended 60-inch width and a minimum width of 48 inches (see PROWAG R303.4 Continuous Clear Width).

The PROWAG limits running slope (slope in the direction of travel) for all ramps at 1:12 or 8.3% and cross slope (slope perpendicular to travel) at 1:48 or 2.1% (PROWAG R304.2.1 Running Slope and R304.2.2 Cross Slope).

Detectable warning surfaces shall be provided in accordance with the PROWAG (PROWAG R304.1 General and R205 Detectable Warning Surfaces). If the temporary ramp provides access to an enclosed safety corridor for passage along the construction zone, accessible truncated domes must be provided at the location where a street crossing occurs. If the temporary ramps do not lead to an area where pedestrians may conjoin the vehicular traffic lane or bike lane, detectable warnings may not be required. These situations are typical where construction is taking place in smaller patch-work areas along sidewalks and pedestrians are diverted into the roadway in a completely protected area with two curb ramps providing access.

Some agencies have established specific requirements for the design of temporary curb ramps to accommodate pedestrians with mobility disabilities. Many private vendors now manufacture and sell temporary curb ramps for use at vertical curbs and ramps for use to access entrances at businesses. These elements are durable and adjustable to conditions on-site. These elements can be moved from construction site to site and may be more cost effective than other options.

Crosswalks in TPAR

Cross slopes at crosswalks are regulated based on intersection control or where it is located, such as at roundabouts or midblock crossings. The cross slope at crosswalks that are not controlled or that are controlled

by a traffic signal, pedestrian hybrid beacon, RRFB, or similar pedestrian signal can match the running slope of the roadway up to a maximum of 5% (PROWAG R302.5.2.2 Crosswalk at Uncontrolled Approach and R302.5.2.3 Crosswalk with Traffic Control Signal or Pedestrian Hybrid Beacon). The cross slope at intersections that are controlled by a stop or yield sign cannot exceed 2.1% (PROWAG R302.5.2.1 Crosswalk with Yield or Stop Control Devices). The cross slope at intersections at midblock crosswalks or roundabouts cannot exceed the existing slope of the roadway (PROWAG R302.5.2.4 Midblock and Roundabout Crosswalks). Temporary crosswalk markings should be installed, when appropriate, to identify where pedestrians will be crossing the street and to improve yielding compliance by other road users.

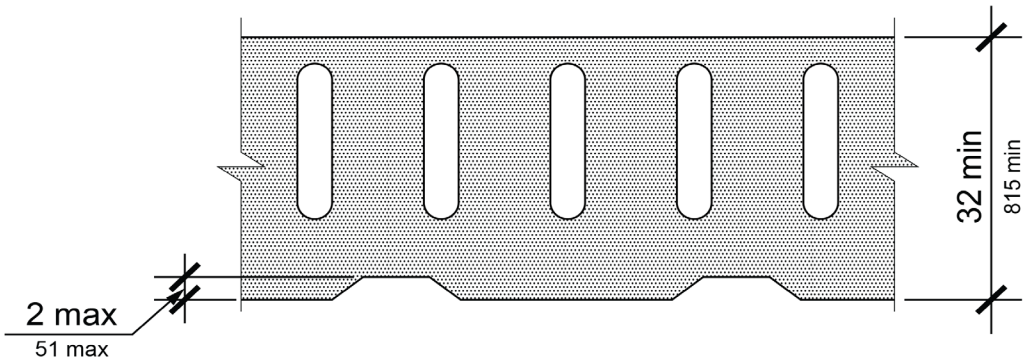
Lighting

Lighting is important to all pedestrians, not only enabling them to see signs and instructions clearly but also to make pedestrians visible to automotive traffic, bike traffic, and other pedestrians. Consideration should be given to avoid trees and other objects that would cast a shadow and obstruct the light output. A TPAR is required to remain day and night with existing barriers and signs in place to protect and inform pedestrians (see MUTCD Section 2A.21 Retroreflection and Illumination, Standard 03-05 and Section 6F.01 General Characteristics of TTC Zone Signs, Standard 11-12).

Detectable Edging

A pedestrian with a vision disability experiences physical barriers daily. Their safety depends on knowing their intended path of travel and not diverting from it, unless they are forced to do so. Signs, drums, cones, caution tape, bollards, or typical type I, II, or III barricades themselves do not provide sufficient guidance to pedestrians with vision disabilities.

Where a channelizing device is used to delineate an TPAR, continuous detectable edging shall be provided throughout the length of the route. Where pedestrians or vehicles turn or cross, gaps in the detectable edging are permitted (PROWAG R303.6 Detectable Edging of Channelizing Devices). These barricades must be at least 32 inches high and have a bottom edge that is continuous, no higher than 2 inches from the walking surface, and detectable by a white cane (see PROWAG Figure R303.6 below).



PROWAG Figure R303.6 Detectable edging of channelizing devices

Joints between channelizing devices should be closed and flush to prevent canes or small wheels from being trapped, to reduce the risk of tripping, and should be free of sharp or rough edges or objects that may harm hands, arms, or clothing. Furthermore, when used as a sidewalk closure mechanism, channelizing devices must run the entire width of the sidewalk; gaps in coverage could lead to people with vision disabilities accidentally crossing into the work zone, where they may be exposed to trips, falls, and severe injuries. The

PROWAG requires (and the MUTCD supports) that no part of the device used may extend more than 4 inches into the pedestrian pathway because it could cause a wheelchair or motorized cart to get stuck or overturn, could trip a person with a vision disability, or could cause someone using a cane to become disoriented (see PROWAG R402.2 Protrusion Limits and MUTCD Section 6C.02 Pedestrian Considerations, Guidance 11(G)).

Audible Messaging

Signs identifying TPAR shall be provided in advance of decision points. Proximity actuated audible signs or other non-visual means of conveying the information that identifies the TPAR shall also be provided (PROWAG R303.2 Signs).

Commercially available AIDs can provide the required audible signs using proximity actuated infrared or other motion sensors. The audible message can be customized so the devices can be reused for multiple construction projects over time. Since AIDs are known to be susceptible to theft, the devices should be secured to the traffic control device (barricade, channelization, fence, etc.).

There are a variety of audible devices available that provide information based on proximity sensors (see photo). Appropriate device choice may be site-specific and should be included in the approved TPAR plan.



Source: SZS Engineering, Inc.

One of the considerations for the work zone is the setup and recording of the audible message. This is site-specific to ensure proper messaging for pedestrians. The audible message should be brief, to the point, spoken clearly, and recorded in a quiet area. Potentially, record a message in more than one language.

Specific message elements to include:

- Brief announcement to get pedestrian's attention.
- Current location of the pedestrian.
- Directions that do not require vision.
- Use of cardinal directions.
- Description of the TPAR in the number of blocks to be traversed or landmarks to be passed.
- Description of conditions where the TPAR will traverse major garage or retail driveways, as these can be mistaken as intersections by people with vision disabilities.
- Duration (if accurate duration is known).

Message example: “Attention southbound Lyndale Avenue pedestrians. You are at the southwest corner of Lyndale and Franklin. West sidewalk closed from 22nd to 26th street for 4 blocks. Cross Lyndale for more bypassing messages. This detour will continue until Friday, May 16.” [20]

Currently, there is no specific federal guidance specifying what information should be included in the speech messages for AIDs. A project funded by the Federal Highway Administration is underway to develop model audible information messages to be used when sidewalk closures are required in work zones.

Physical Separation

TPAR plans should be designed to minimize conflicts between vehicular and pedestrian movements. Consideration should be made to separate pedestrian movements from worksite activity and vehicular traffic. Unless an acceptable route that does not involve crossing the roadway can be provided, pedestrians should be appropriately directed with advance signing that encourages them to cross to the opposite side of the roadway. In areas with high vehicular traffic volumes, signs should be placed at intersections (rather than midblock locations) so that pedestrians are not confronted with midblock worksites that could induce them to attempt to navigate through the worksite or make a midblock crossing (MUTCD Section 6C.02 Pedestrian Considerations, Guidance 09). Where possible, positive physical separation should be used when pedestrians are moved off the sidewalk and TPAR walkways are provided in parking areas, bicycle lanes, or the roadway. The positive separation between vehicles and pedestrians that will be used should be specified in the written TPAR plan.

Safe TPAR walkways can use different materials based on traffic volume. Higher-volume or higher-speed roadways will require more intensive, crash rated, separation devices such as water filled longitudinal barriers, concrete barriers, etc., whereas longitudinal channelizing devices may be acceptable on lower speed and volume roadways. In any case, drums, bollards, caution tape and cones, or similar will not provide adequate separation or continuous cane detectable warning for pedestrians traveling through a work zone adjacent to a roadway, drive lane, or parking area (see PROWAG R303.6 Detectable Edging of Channelizing Devices).

Color contrast is important for pedestrians with low vision and should be considered when selecting channelizing devices. Orange is the standard color per the MUTCD for TTC as it provides adequate color





Source: www.concretesafetysystems.com

[20] Minnesota Department of Transportation, Temporary Pedestrian Access Routes (TPAR) Audible Message Content (January 2018)

contrast and seldom results in glare (MUTCD Section 1D.05 Color Code, Standard 02(I)). Do not use black to color any base on a device. Black can blend in with the asphalt walkway surface and could cause a tripping hazard. The bottom and top surfaces of devices should have retroreflective material or delineation on the side exposed to traffic for improved nighttime visibility.

v. TPAR Implementation

Each work zone is different and will require TPAR that suits the needs of the conditions on site. Common issues that adversely affect pedestrians at or near work zones can be found in each type of work zone, although the solution may vary depending on several factors. When considering the type of TPAR to best suit the needs of pedestrians, three common sets of conditions exist where pedestrian modifications at work zones are necessary:

1. TPAR within Work Zone
2. TPAR Diverted Around a Work Zone
3. TPAR Detour/Diversion

Pedestrian travel must be a consideration when planning the staging of construction phases to ensure that pedestrians can travel safely, especially where more than one parallel path of travel exists. Both paths of travel should not be closed at the same time, unless an incident such as an emergency exists. With proper planning, at least one path of travel serving the public should remain open.

1. TPAR within Work Zone

One common situation that pedestrians encounter is work zones located in the roadway or on property immediately adjacent to a path of travel or sidewalk. Even where the actual path of travel or sidewalk is not being worked on or is not formally closed, work zones adjacent to the pedestrian path can present hazards and impediments to pedestrians. When doing this type of work, field personnel must take measures to safeguard pedestrians. Caution tape and wooden stakes are not cane detectable warnings that the blind or visually impaired can recognize.



Source: SZS Engineering, Inc.

Protection for Pedestrians at Work Zones Adjacent to the Path of Travel or Sidewalk

In low volume, low speed residential areas it may be possible for the sidewalk to remain open adjacent to a work zone. If there is not 48 inches of clearance adjacent to the work zone, detectable and continuous edge protection must be provided to allow pedestrians with a vision disability to safely navigate work zones. Even with sufficient clearance, pedestrians should be protected from excavations and unexpected vertical drop-offs using proper protection and channelization devices. Continuous longitudinal channelizers and properly constructed pedestrian barricades can safely separate pedestrians from work zones and provide positive path guidance for the blind and visually impaired.

Pedestrian Protection – Work Zone Safety

If hazards exist within the work zone that require safety precautions for workers (trenches, fall protection, hard hats, crane operations, etc.), protection must be provided to ensure pedestrians do not accidentally or purposely walk into and across the work zone.

Continuous concrete barriers, plastic longitudinal channelizing devices, wooden enclosed walkways, or cyclone fencing with a continuous, smooth edge provided at the bottom can be used for this purpose. A cane detectable warning is required along these elements that does not create a hazard for other pedestrians. For instance, the bottom edge of the fence serves the blind as a cane detectable guide for safe travel, but when the footing protrudes into the walking surface, it becomes a trip hazard for other pedestrians (see photo). These types of protrusions and abrupt changes must be avoided to ensure all pedestrians have a safe and usable walking surface to travel along.



Source: SZS Engineering, Inc.

Pedestrian and Motorist Lines of Sight

Devices used to prohibit pedestrian access or provide protection from hazards can impede pedestrians and motorists from seeing each other or important traffic control devices if not designed or installed properly. Similarly, the improper positioning of TTC devices for motorists may also impede their ability to see pedestrians and should be moved. The location of work vehicles, equipment, and materials should be monitored to maintain maximum visibility.

End of Shift – Obstacles

At all times, the pedestrian path of travel or TPAR must be left clear of mud or dirt, poles, temporary signs, hoses, barriers, construction materials, vehicles, equipment, and other protruding objects such as tree limbs that are part of the work zone and could present hazards to any pedestrian. The TPAR should be reviewed as part of this process daily to ensure it remains in place and functions as intended.

Objects with leading edges more than 27 inches and less than 80 inches above the walking surface shall not protrude horizontally more than 4 inches into pedestrian circulation paths. Handrails shall be permitted to protrude 4 ½ inches maximum (PROWAG R402.2 Protrusion Limits). Where objects are mounted on a single post or pylon and the objects are more than 27 inches and less than 80 inches above the walking surface, the objects shall not protrude into the pedestrian circulation path more than 4 inches measured horizontally from the post or pylon or more than 4 inches measured horizontally from the outside edge of the base where the base height is 2 ½ inches minimum (PROWAG R402.3.1 Objects Mounted on Single Post or Pylon).

Head room minimum height must also be maintained throughout the TPAR. Where the TPAR uses a canopied walkway to protect pedestrians from falling debris or signs are installed within the pedestrian path, the MUTCD states that such a walkway or sign should provide at least 7 feet (84 inches) of headroom (see MUTCD Section 6F.02 Sign Placement, Standard 05-06), although the ADAS and the PROWAG require only 80 inches (PROWAG R402.4 Vertical Clearance and ADAS 307.4 Vertical Clearance).

2. TPAR Diverted Around a Work Zone

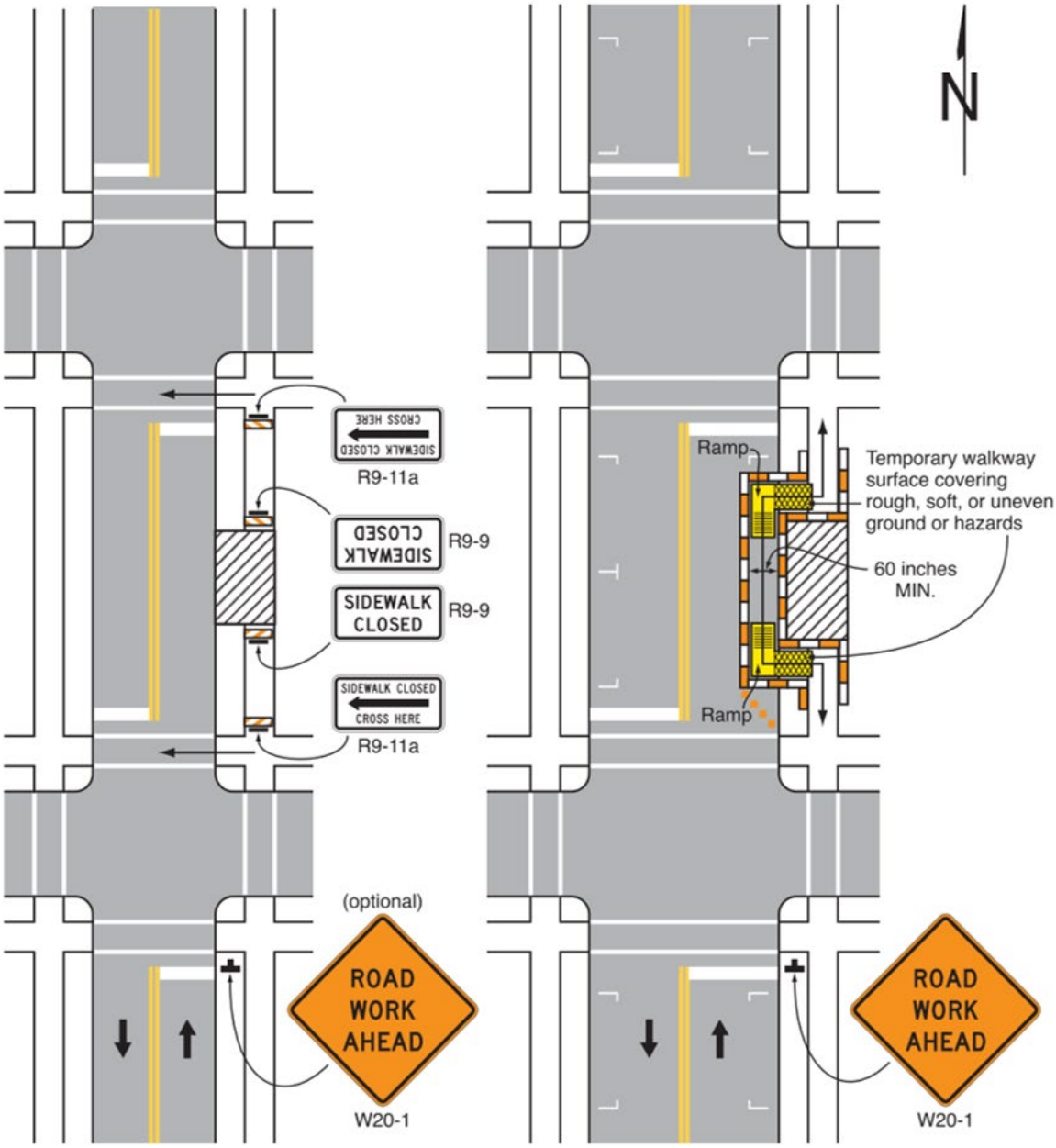
Most work activities block a sidewalk or path for several days. Unless workers are stationed to assist pedestrians around the blockage day and night, a temporary diversion path or TPAR will be required to allow pedestrians to safely travel around the blockage.

Speed determines the type of separation required at work zones. In areas where the speed is less than 30 mph, it is advisable to maintain longitudinal separation with continuous channelization devices, curbing, railings, or barricades at a minimum. If the work zone is in an area of high crash frequency or if there is concern over pedestrian exposure and risk, the use of positive protection should be considered. If the work zone is in an area where the speed is 30 mph or more, positive protection shall be used. All positive protection should be installed and used following the manufacturer's specifications. Alternatively, consider detouring pedestrians across the street or around the block to open, permanent pedestrian facilities, as needed.

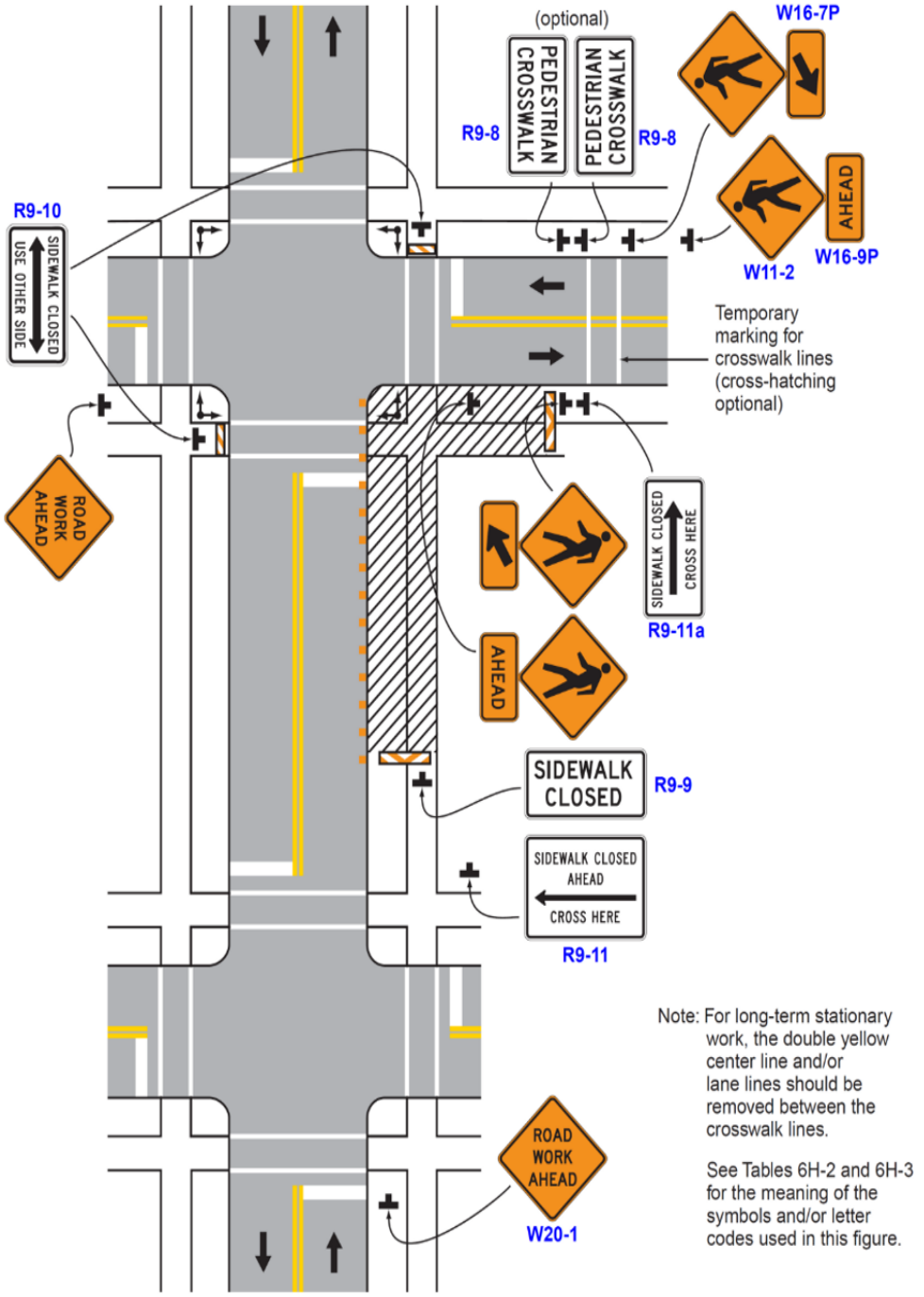
3. TPAR Detour/Diversion

The third category of pedestrian modification requires establishing more complex detour routes for pedestrians onto other walking surfaces that include sidewalks, paths of travel, or other options. In many ways, these pedestrian detours are similar to the TPAR required at road closures. The PROWAG and the MUTCD provide guidance on signs and barricades needed to detour pedestrians to alternate sidewalks or paths. It is not only time consuming for pedestrians with disabilities to be forced to retrace their steps when they reach a dead-end at a work zone, but it can also be unsafe. TPAR must be installed and maintained to ensure that pedestrians are diverted far enough upstream to utilize the detour as part of typical travel.

See the two examples from the MUTCD (Figures 6P-28 and 6P-29) below, providing guidance on TPAR sign placement and detour design:



MUTCD Figure 6P-28. Sidewalk Detour or Diversion (TA-28)



MUTCD Figure 6P-29. Crosswalk Closures and Pedestrian Detours (TA-29)

Barricades and barriers used to close the sidewalk or path of travel must be properly installed and maintained. The appropriate signs, markings, channelization, and barriers should be specified in the TPAR plan, including the identification of locations of necessary temporary ramps and audible pedestrian warning devices to accommodate pedestrians with mobility or vision disabilities.



Source: SZS Engineering, Inc.

Design and Install Pedestrian Detour Signing and Channelization Appropriately

Pedestrian detours should be part of the written TPAR plan so that all field personnel share the same information. Signing should be installed to warn pedestrians of downstream sidewalk/path closures and to provide information as to where the detour path is located.

Where possible, it is important to inform pedestrians about what businesses are still accessible along the sidewalk or path. Signs should conform to the examples shown below, from MUTCD Figure 6G-1. Hand drawn signs or messages are not sufficient for guiding pedestrians.



MUTCD Figure 6G-1. Regulatory Signs and Plaques in Temporary Traffic Control Zones (image excerpt)

TPAR Inspection & Maintenance

When signs and channelization for the pedestrian detour route are installed, the route should be checked on a regular basis to make sure it will accommodate all users.



Source: SZS Engineering, Inc.

The TPAR route shall:

- Match the approved/permitted plan set.
- Be continuous with no gaps between crosswalks and channelizers or barricades (exception: where pedestrians or vehicles turn or cross).
- Not have equipment or parking in the walking path.
- Have protection from trenches and holes.
- Maintain visibility for users.
- Not have mud, dirt, or debris on walking surfaces (asphalt surfaces to be smooth).

- Not have signs, hoses, barriers, materials, vehicles, equipment, or protruding objects that prohibit passage.
- Have compliant temporary ramps.
- Ensure that audible messages are updated accordingly as changes to the TPAR occur.

VI. Delegation of Authority & Eugene City Code – Enforcement

Chapter 7 of the Eugene City Code covers public improvements, including construction requirements for individuals, contractors, and franchisees, and permits for use of public way. The following sections concern delegation of authority to the City Engineer, permits, and enforcement:

- Eugene City Code 7.007 Rules and Regulations
- Eugene City Code 7.290 Public Way Construction and Use – Permit Required – Standard Specifications, Drawings and Design Standards
- Eugene City Code 7.295 Public Way Construction and Use – Permit Application
- Eugene City Code 7.299 Public Way Construction and Use – Permit Suspension or Revocation

VII. Reference List: TPAR Requirements and Sources

The following is a list of TPAR requirements with citations to the current versions of the affiliated sources for additional information and reference. This list is derived from Appendix B of the General Release and Settlement Agreement in Moore, et al. v. City of Eugene, et al.[21]

Before any new sidewalk closure or detour is active, all necessary signs and other accessibility features shall be in place.

- MUTCD: Section 1D.09 Standard 01 and Option 02
- PROWAG: R204 and R205
- ODOT TTCH: Section 1.5
- COE Manuals: PIDS 3.05; SSC 220.02

Provide a pedestrian access route during construction that is comparable to pre-construction conditions.

- MUTCD: Sections 6C.02 Standard 03, 6C.02 Guidance 07(C), and 6C.03 Standard 04
- PROWAG: R204 and R303
- ODOT TTCH: Sections 1.5 and 5.2
- COE SD: TM 844

[21] No. 6:21-cv-01293-MTK, (D. Or. filed Aug. 31, 2021)

- COE Manuals: PIDS 3.05; SSC 220.02

Provide non-slip PAR of minimum 60-inches wide, no greater than 2% cross slope (48 inches with documented design exception).

- PROWAG: R302
- ODOT TTCH: Section 5.2
- COE SD: TM 844
- COE Manuals: SSC 220.02 and 220.60

Ensure ramps on detour route are comparable; include compliant temporary ramps that meet PAR requirements and include detectable warnings.

- MUTCD: Sections 6C.02 Standard 03, 6C.02 Guidance 07(C), and 6C.03 Standard 04
- PROWAG: R303 and R407
- ODOT TTCH: Section 5.2
- COE SD: TM 844
- COE Manuals: PIDS 3.05; SSC 220.02, 228.13, and 228.43

Provide PAR and pedestrian circulation path clear of obstructions and protruding objects.

- PROWAG: R204 and R210
- ODOT TTCH: Section 2.5
- COE Manuals: PIDS 2.03; SSC 220.02

Provide cane detectable pedestrian channelization & barricades: At sidewalk closures, barricades must be full width of sidewalk and extend to within 2 inches of the ground; must separate pedestrians from traffic; cannot use cones or caution tape for barricades.

- MUTCD: Sections 6C.02 Guidance 28, 6C.03 Support 05, and 6M.04
- PROWAG: R204, R303, and R303.6
- ODOT TTCH: Section 5.2
- COE Manuals: PIDS 3.05; SSC 220.02 and 228.10

Protective and cane detectable barriers around open trenches and drop-offs.

- PROWAG: R303.6
- COE SD: TM 844 and TM 820
- ODOT TTCH: Section 5.2

- COE Manuals: SSC 220.02

Ensure push buttons are operable and accessible; any temporary buttons must be accessible.

- MUTCD: Sections 6C.02 Standard 03, 6C.03 Standard 04, 4I.05, and 4K.01-05
- PROWAG: R204, R206.3, R303.7, and R403
- COE Manuals: PIDS 2.03 and 3.05; SSC 227.12 and 227.13

Sidewalk closed warning signs in advance of closure to alert people of detour route.

- MUTCD: Section 6C.02 Guidance 02
- PROWAG: R303.2
- ODOT TTCH: Section 5.2
- COE SD: TM 844
- COE Manuals: PIDS 3.05; SSC 220.02

Limit curb ramp work to one corner at a time, unless on-site detour routes are maintained.

- MUTCD: Section 6N.04 Standard 08-09
- COE SD: TM 844
- COE Manuals: PIDS 3.05; SSC 220.02

Minimize out-of-direction travel and avoid detours/crossings.

- MUTCD: Section 6C.02 Support 06 and 6N.04 Guidance 07
- COE SD: TM 844
- COE Manuals: PIDS 3.05; SSC 220.02

Provide audible information devices for pedestrians with vision disabilities.

- MUTCD: Section 6C.03 Guidance 06 and Support 07
- PROWAG: R303.2
- COE Manuals: PIDS 3.05

All TTC devices shall be removed as soon as practical from sidewalks and sidewalk detour routes.

- MUTCD: Section 6A.03 Standard 04
- ODOT TTCH: Section 6.3 and Appendix C
- COE Manuals: PIDS 3.05

When transit stops are affected or relocated, pedestrian access to the affected or relocated transit stops shall be provided.

- MUTCD: Sections 6C.02 Standard 03, 6C.03 Standard 04, and 6C.02 Guidance 11(B)
- PROWAG: R204.2 and R309.1.3.2

Curb parking prohibited for at least 50 feet in advance of midblock crosswalks.

- MUTCD: Section 6P.01 (Notes for Figure 6P-29) Standard 2

Appendix: List of Figures

PROWAG Figure R303.6 Detectable edging of channelizing devices

MUTCD Figure 6P-28. Sidewalk Detour or Diversion (TA-28)

MUTCD Figure 6P-29. Crosswalk Closures and Pedestrian Detours (TA-29)

MUTCD Figure 6G-1. Regulatory Signs and Plaques in Temporary Traffic Control Zones